ORGNZ-09: Enterprise Data Organization Patterns

> **Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 9 of 10 | **Created:** January 2026 | **Last Updated:** 11/07/2026

Overview

This notebook brings together buckets, segments, and security context into comprehensive enterprise patterns. Learn how to combine these mechanisms for effective data governance at scale.

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	`storage:bucket-definitions:read`, `storage:logs:read`, `storage:filter-segments:read`
Knowledge	Completed ORGNZ-08 (Grail Segments)
Data	At least 1 hour of log and entity data

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Learning Objectives

By the end of this notebook, you will:
- Combine buckets, segments, and security context effectively
- Design enterprise-scale data organization architectures
- Implement common organizational patterns
- Plan data governance strategies

Combining All Three Mechanisms

For comprehensive data governance, use all three mechanisms:

Mechanism	Purpose	When to Use
Buckets	Physical storage, retention	Different retention or access needs (cost attribution uses `dt.cost.*` fields)
Segments	Logical filtering, views	Team-focused data views, cross-signal filtering
Security Context	Access enforcement	Fine-grained security requirements

Complete Architecture Example

```
Finance Team Requirements:

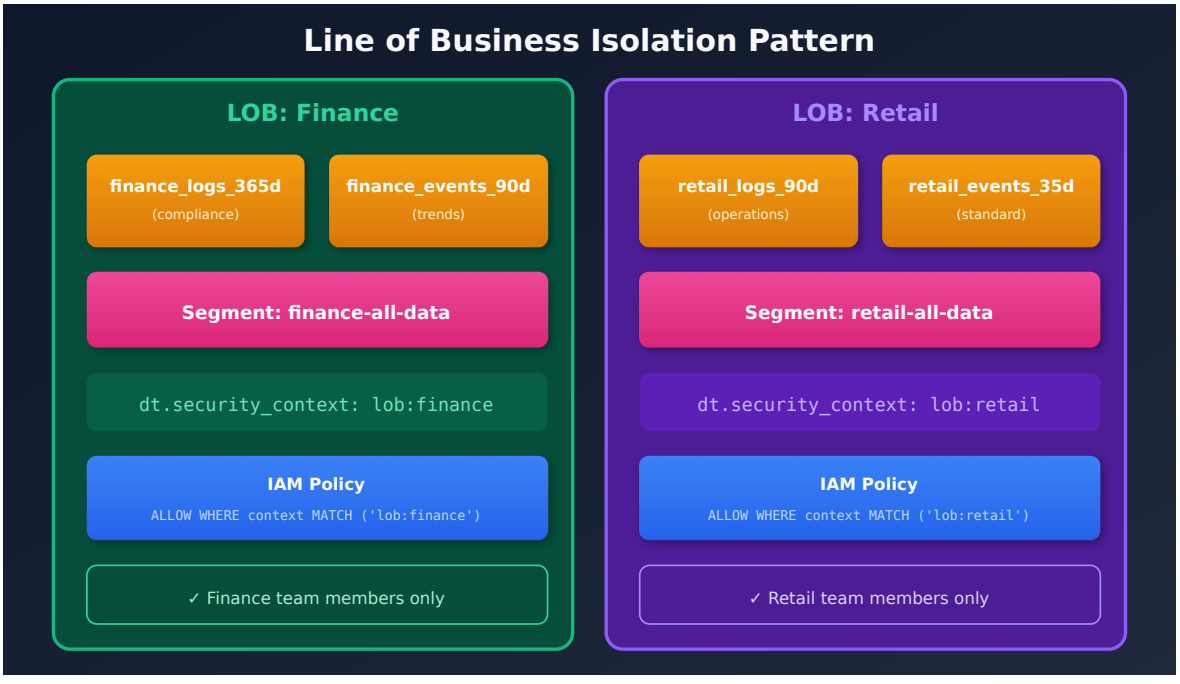
Bucket: finance_logs_365d
  - 1 year retention for compliance
  - Cost attributed with dt.cost.costcenter set at the source (not by the
bucket)

Segment: finance-team-view
  - Filtered view of finance data
  - Includes related services and hosts

Security Context: lob:finance
  - IAM policy: ALLOW WHERE security_context MATCH ('lob:finance')
  - Only finance team members can access
```

Enterprise Pattern 1: Line of Business Isolation

Complete isolation by business unit:



Implementation

Step 1: Create buckets

```
finance_logs_365d (logs, 365 days)
finance_events_90d (events, 90 days)
retail_logs_90d (logs, 90 days)
```

Step 2: Assign buckets in OpenPipeline

In the log pipeline's **Storage** stage, add one **Bucket assignment** processor per LOB; the first matching processor wins (see **ORGNZ-03 § Routing Data to Buckets**):

Matching condition (DQL)	Bucket
`matchesValue(dt.host_group.id, "finance-*")`	`finance_logs_365d`
`matchesValue(dt.host_group.id, "retail-*")`	`retail_logs_90d`

Step 3: Set security context

Set `lob:finance` / `lob:retail` at the source first — a `dt.security_context=lob:finance` host tag on the finance hosts, or a fixed value per host group in the central primary Grail tag configuration (see **ORGNZ-06 §3**). Where the value has to be computed in the pipeline, use **Set security context** processors in the pipeline's Permission stage, which *"Sets the proper record-level access via dt.security_context by either copying it from a field, setting it as a static string, or a static array that allows multiple values."*

Matching condition (DQL)	Security context (static string)
`matchesValue(dt.host_group.id, "finance-*")`	`lob:finance`
`matchesValue(dt.host_group.id, "retail-*")`	`lob:retail`

> **Corrected 09/24/2026.** Earlier versions of this notebook showed `type: route` and `type: security-context` YAML processors keyed on `host.group`. Neither processor type exists in that form, and `host.group` is not a Grail field — use `dt.host_group.id`.
>
> **Sources:** Processing in OpenPipeline (DT docs), DQL matcher in OpenPipeline (DT docs).

Step 4: Create IAM policies

```
// Finance policy
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "finance_";
ALLOW storage:logs:read, storage:events:read WHERE storage:dt.security_context
MATCH ("lob:finance");
```

> **Use `MATCH`, not `=`, for events.** `dt.security_context` on events is an array — on the validation tenant every event carrying a security context held an array (522,703 events over 7 days, checked 11/07/2026) — and *"Using `=`, `STARTSWITH` or `IN` when the field holds an array will always return `false`."* With `=`, the events half of this statement grants nothing and nothing errors. `MATCH` matches a single string or any element of an array, so one statement covers logs and events.
>
> **Sources:** Permissions in Grail (DT docs) — *"If you expect your record filters might contain an array, use the MATCH operator in your IAM statements."*

Enterprise Pattern 2: Environment-Based Tiering

Different access levels for production vs non-production:

```
Production:
├── Buckets: prod_logs, prod_events, prod_spans
├── Security Context: env:production
├── Policy: ALLOW WHERE storage:dt.security_context = 'env:production'
├── Access: Restricted to production support team
└── Audit: Full audit logging enabled

Non-Production:
├── Buckets: default_logs, default_metrics, default_spans
├── Security Context: env:dev, env:staging, env:qa
├── Policy: ALLOW WHERE storage:dt.security_context MATCH ('env:dev*',
'env:staging*')
├── Access: Broader developer access
└── Audit: Reduced audit requirements
```

Tiered Access Groups

Group	Production Access	Non-Prod Access
SRE Team	Full	Full

Group	Production Access	Non-Prod Access
Production Support	Read-only	None
Developers	None	Full
QA	None	Read-only

Enterprise Pattern 3: Multi-Cloud Organization

Organize by cloud provider and account:

```
AWS:
├── Bucket: aws_logs_35d
├── Security Context: cloud:aws/&lt;account-id&gt;
└── Policy: ALLOW WHERE storage:aws.account.id = '123456789012'

Azure:
├── Bucket: azure_logs_35d
├── Security Context: cloud:azure/&lt;subscription&gt;
└── Policy: ALLOW WHERE storage:azure.subscription = 'sub-id'

GCP:
├── Bucket: gcp_logs_35d
├── Security Context: cloud:gcp/&lt;project&gt;
└── Policy: ALLOW WHERE storage:gcp.project.id = 'my-project'
```

Enterprise Pattern 4: Kubernetes-Native Organization

Leverage Kubernetes constructs for organization:

```
Cluster: main-cluster
├── Namespace: production
|   ├── Security Context: k8s:main-cluster/production
|   └── Policy: ALLOW WHERE storage:k8s.namespace.name = 'production'
|
├── Namespace: staging
|   ├── Security Context: k8s:main-cluster/staging
|   └── Policy: ALLOW WHERE storage:k8s.namespace.name = 'staging'
|
└── Namespace: development
    ├── Security Context: k8s:main-cluster/development
    └── Policy: ALLOW WHERE storage:k8s.namespace.name = 'development'
```

Cross-Namespace Segment

One application deployed to several namespaces, as one segment. Build it in the Segments app with a single include:

Include (data object)	Filter
Data (all types)	`k8s.workload.name = "checkout"`

Segment conditions take only `=` and `in()` — *"Only operators `=`, and `in()` are supported."* — with `*` wildcards inside the value, so write `=` and double quotes, not DQL's `==`. One `Data (all types)` include replaces separate `logs` and `spans` includes. Check coverage first: `k8s.workload.name` is a `stable` semantic-dictionary field, but on the validation tenant on 10/14/2026 it was on 41% of logs in an hour, and records without it fall outside the segment (adapt the coverage audit in ORGNZ-10 §9).

> **Sources:** Segment limits (DT docs) — as quoted, Supported data types in segments (DT docs) — *"We recommend constructing segments with a single Data (all types) include block when possible."*
>
> **Dictionary:** `k8s.workload.name` (`stable`), read 10/14/2026.

Decision Framework

Use this framework to choose your organization strategy:



Implementation Checklist

Phase 1: Planning

- [ ] Identified retention requirements
- [ ] Mapped organizational structure (teams, LOBs, cost centers)
- [ ] Defined access control requirements
- [ ] Planned bucket naming convention
- [ ] Designed security context schema

Phase 2: Infrastructure

- [ ] Created custom buckets
- [ ] Configured OpenPipeline bucket assignment (Storage stage)
- [ ] Configured OpenPipeline security context processors
- [ ] Verified data flow to correct buckets

Phase 3: Access Control

- [ ] Created IAM policies
- [ ] Assigned policies to groups
- [ ] Tested access with sample users
- [ ] Documented all policies

Phase 4: Views

- [ ] Created segments for team views
- [ ] Tested segment filters
- [ ] Documented segment purposes

Phase 5: Governance

- [ ] Planned access review process
- [ ] Configured audit logging
- [ ] Created runbook for access management

Auditing Your Organization

DQL: Environment Audit

Measure security context adoption across all ingested logs:

```
// Audit security context coverage — percentage of logs with
dt.security_context assigned
fetch logs, from:-1h
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    withoutContext = countIf(isNull(dt.security_context))
| fieldsAdd coverage = round(toDouble(withContext) / toDouble(total) * 100,
decimals: 2)
```

Best Practices Summary

Area	Best Practice
Buckets	Target 1–3 TB/day per log bucket, a dedicated bucket per application above 250 GB/day (ORGNZ-02); use consistent naming
Segments	Test rules with DQL before deploying
Security Context	Use hierarchical encoding for flexibility
IAM Policies	Assign to groups, not individuals
Documentation	Document all policies and contexts
Reviews	Schedule regular access reviews

Series Summary

The ORGNZ series covered:

Notebook	Topic	Key Takeaway
ORGNZ-01	Introduction	Why organize data; three pillars
ORGNZ-02	Grail Buckets	Bucket fundamentals and limits
ORGNZ-03	Bucket Strategy	Naming, retention, design patterns
ORGNZ-04	Permissions Overview	Permission levels and policy structure
ORGNZ-05	Bucket-Level Access	IAM policies for bucket isolation
ORGNZ-06	Security Context	Fine-grained access with dt.security_context
ORGNZ-07	Advanced Permissions	Record and field-level patterns

Notebook	Topic	Key Takeaway
ORGNZ-08	Grail Segments	Dynamic data organization
ORGNZ-09	Enterprise Patterns	Combined approaches at scale
ORGNZ-10	Advanced Segment Definitions	Filter syntax, enrichment, variables, troubleshooting

References

- Organize data
- Permissions in Grail
- Advanced permission setup
- Enhance data management with Grail
- Set up Cost Allocation (DT docs)

This notebook was AI-generated from Dynatrace documentation and enterprise best practices. It is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.